A-A-54141
20 April 1990

COMMERCIAL ITEM DESCRIPTION

ANALYZER, INTRAVENOUS INFUSION PUMP

The General Services Administration has authorized the use of this commercial item description.

This commercial item description covers an intravenous infusion pump analyzer.

Salient characteristics.

The analyzer shall be a portable unit suitable for testing the flow-rate of intravenous infusion (IV) pumps and similar volumetric hospital equipment.

The unit shall contain a microprocessor which controls the tests and calculates and displays the results. Shall be supplied with two volumetric chambers, one large and one small, each with accuracy of one (1) percent. The unit shall have modular solid state circuitry and digital read outs. The unit shall be of modular construction. Shall have subassemblies and chassis components which are removable without removing any other hard wired assembly or printed circuit card or component. Shall have printed circuit boards which are removable without the need to unsolder cables or wires. The unit shall permit adjustments without removing any component printed circuit boards, or modules. The use of a circuit board extender to permit adjustments is acceptable. The unit shall be capable of testing four IV infusion pumps simultaneously.

Operation. The unit shall be capable of performing both a drop rate test and a volumetric flow rate test. The unit shall also be capable of converting drop rate to volumetric output. Selection switches shall describe function, range, and/or test.

AMSC/NA FSC 6515

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Volumetric test. The unit shall calculate volumetric infusion pump output flow in milliliters per hour by measuring the time required to fill a volumetric chamber. The unit shall measure and display volumetric flow rate in milliliters per hour. The analyzer shall be capable of performing two simultaneous tests independently. Shall have digital readouts with ranges of 0 to 9999 ml/hr, and 0 to 999.9 ml/hr.

Drop rate test. The analyzer shall be capable of calculating the average drop rate of drop calibrated infusion pumps in drops per minute by counting the drops sensed by a photocell probe and keeping track of the total test time. The unit shall be capable of performing two simultaneous tests independently. The analyzer shall control the test automatically upon initiation by a switch. Shall keep track of three parameters: drop rate average (per minute), drop deviation (mean), and drop test time (minutes). The digital readout shall display drop rate and drop deviation in the following range: 0 to 250 drops/minute and test time 0 to 9999 minutes.

Controls. All controls shall be located on the front panel. Selection switches and terminals shall describe function range and/or test.

Displays. The unit shall have digital readouts consisting of at least four (4) digits and be capable of displaying average drop rate, minute to minute drop count, mean deviation, test time, average mean deviation, and equivalent ml/hr. The displays may be light emitting diodes or liquid crystal.

Alarms. The analyzer shall be provided with alarms to indicate: zero drop rate, drop rate variation, and test complete.

Accuracy. The analyzer shall have an accuracy falling within the following tolerances: for timing, plus or minus 200 milliseconds. For collecting volume, plus or minus 1 percent, for display, plus or minus 1/2 digit.

Environmental. The analyzer shall perform as specified within the following conditions: storage temperature -46° C to plus 71° C. Operate at temperatures 0° C to plus 55° C. Humidity of 95 percent, and environment of 30 mppcf (10 mg/m^3) of total dust with less than 1 percent quartz or 5 mg/m^3 respirable dust.

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Dimensions. The unit shall have overall dimensions which do not exceed 500 cubic inches and shall weigh no more than seven (7) pounds.

Electrical. The unit shall operate on a nominal 115/230 volt single phase 50/60 Hz power source. The unit shall comply with applicable requirements of UL 1244, measuring and test equipment.

Maintenance. The unit shall be designed for a service life of at least seven (7) years. The unit shall be supplied with organizational maintenance devices (fuses, batteries, etc.) which are replaceable without disassembly of the basic unit.

Workmanship. The analyzer shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One analyzer, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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Sampling for tests. Sampling for tests (dimensional), shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of II. The unit of product for test purposes shall be one intravenous infusion pump analyzer.

Tests. Testing shall be conducted to determine compliance with specified requirements. Testing shall include but not be limited to the following:

Verify the unit is operational on each of the four specified input powers. All remaining tests are to be performed using each specified input power.

Verify the unit has an accuracy within the following tolerance levels:

Timing $\pm$ 200 milliseconds

Collecting volume $\pm$ 1 percent

Display $\pm$ 1/2 digit

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirement of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specification, standard, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock number is covered by this document:

NSN	ITEM IDENTIFICATION
6515-01-234-6805	ANALYZER, INTRAVENOUS INFUSION PUMP

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

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CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-4675

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-54141	ANALYZER, INTRAVENOUS INFUSION PUMP

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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